

1. Administrative & Study Identification

Full Study Title: A Clinical Study of V940 and Pembrolizumab (MK-3475) in People With Melanoma (V940-012/INTERpath-012) **Short Title/Acronym:** INTERpath-012 **Protocol Number:** V940-012 **NCT Number:** NCT06961006 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** V940 (mRNA-4157) and Pembrolizumab (MK-3475, Trade Name: Keytruda, ATC Code: L01FF02) **Phase of Development:** Phase II **Medical Monitor:** Merck Sharp & Dohme Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate Progression-Free Survival (PFS) in participants receiving V940 plus pembrolizumab versus placebo plus pembrolizumab. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DoR), and to assess the safety and tolerability of the combination therapy.

2.2 Background & Rationale To address the need for improved first-line therapeutic strategies in advanced cutaneous melanoma. While standard-of-care immunotherapy (pembrolizumab) has improved survival in advanced melanoma, many patients still experience disease progression. This study is being conducted to evaluate if the addition of V940, an individualized mRNA-based neoantigen therapy designed to prime the immune system against a patient's specific tumor mutations, can significantly improve clinical outcomes and stop cancer from growing or spreading compared to pembrolizumab alone.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo- and Active-Comparator-Controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration **Trial Status:** Recruiting **Target \$N\$:** Approximately 160 evaluable patients **Number of Sites:** Multicenter (Global, including sites in the US, Australia, and other countries) **Estimated Study Start:** 29-May-2025 **Estimated Primary Completion:** 22-Jul-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with unresectable and histologically confirmed Stage III or IV cutaneous melanoma per AJCC Eighth Edition guidelines. 2. Has been untreated for melanoma (except for prior adjuvant or neoadjuvant targeted therapy/immunotherapy if relapse did not occur within 12 months after treatment discontinuation). 3. Have the presence of at least 1 measurable lesion by CT or MRI per RECIST 1.1 and provide tumor tissue suitable for next-generation sequencing and biomarker analysis.

4.2 Exclusion Criteria 1. Clinically significant heart failure (NYHA class III or IV) within the past 6 months, or active/history of immunodeficiency/autoimmune conditions (e.g., HIV-infected with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease). 2. Has ocular or mucosal melanoma. 3. Received prior systemic anticancer therapy for melanoma before randomization or received prior radiotherapy within 2 weeks of the start of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Experimental Arm:** V940 (1 mg) every 3 weeks (q3w) for up to 9 doses + Pembrolizumab (400 mg) every 6 weeks (q6w) for up to 17 doses. - **Control Arm:** Placebo every 3 weeks (q3w) for up to 9 doses + Pembrolizumab (400 mg) every 6 weeks (q6w) for up to 17 doses. **Route of Administration:** Intramuscular (IM) injection for V940/Placebo; Intravenous (IV) infusion for Pembrolizumab. **Duration of Treatment:** Up to approximately 27 weeks for V940/Placebo; up to approximately 2 years for Pembrolizumab (or until disease progression or discontinuation).

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care for expected immune-related adverse events. **Prohibited:** Other systemic anti-cancer therapies, live or live-attenuated vaccines within 30 days prior to the first dose, and other universal or personalized cancer vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue Collection, BRAF V600 Mutation Testing, Baseline Scans (CT/MRI)
Baseline	Day 0	Randomization, First Dose (V940/Placebo IM + Pembrolizumab IV), Vital Signs
Interim	Weeks 3 to 104	V940/Placebo IM (Q3W up to 9 doses), Pembrolizumab IV (Q6W up to 17 doses), Efficacy Assessments (RECIST 1.1)
End of Study	Up to Year 6	Final

Treatment Visit, End of Treatment Safety check, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) - Time from randomization to the first documented disease progression or death from any cause. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) per RECIST 1.1.

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Objective Response Rate (ORR) - Duration of Response (DOR) - Incidence of adverse events (AEs) and safety profile of the combination therapy

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical and laboratory evaluation at every treatment cycle (especially for immune-mediated adverse events and infusion-site reactions). **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An external Independent Data Monitoring Committee (eDMC) will evaluate safety and efficacy continuously during the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses (PFS, OS, ORR); As-Treated/Safety Set for safety parameters. **Sample Size Justification:** \$N \approx 160\$ participants powered to detect a statistically significant improvement in PFS for the V940 + Pembrolizumab arm compared to the Placebo + Pembrolizumab arm. **Handling of Missing Data:** Time-to-event outcomes (e.g., PFS, OS) will be evaluated using Kaplan-Meier methods, right-censoring patients at the last known alive/progression-free date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of clinical data, strict cold-chain supply monitoring for mRNA-4157, and source data verification. **Audit Trail:** Validated 21 CFR Part 11 compliant Electronic Data Capture (EDC) system with full audit trails and data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{V940-012}} **Posting ID:** {{1030324190384348278}}
Registry Number: {{NCT06961006}} **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}} **Study Title:** {{A Clinical Study of V940 and Pembrolizumab (MK-3475) in People With Melanoma}}
Official Regulatory Title: {{A Phase 2, Randomized, Double-Blind, Placebo- and Active-Comparator-Controlled Clinical Study of V940 (mRNA-4157) Plus Pembrolizumab Versus Placebo Plus Pembrolizumab in Participants With First-Line Advanced Melanoma (INTerpath-012)}}

12. Clinical Framework (Melanoma Specific)

Primary Condition: {{Advanced Malignant Melanoma}} **Disease Codes:** {{MeSH: D008545 | HPO: HP:0002861 | SNOMED: 372244006}} **Diagnosis Threshold:** {{Unresectable Stage III or IV Cutaneous Melanoma}}
Medical Methodology: {{Combination Therapy (Immunotherapy + Individualized Neoantigen Therapy)}} **Optimization Target:** {{Progression-Free Survival and Tumor Shrinkage}}

13. Study Procedures & Methodology

Management Pathway: {{First-line advanced melanoma oncological pathway}} **Education Protocol (HCP):** {{Handling, storage, preparation, and administration guidelines for mRNA vaccines and PD-1 inhibitors}} **Education Protocol (Patient):** {{Onsite training on identifying and reporting immune-mediated adverse events}}
Quality Audit Mechanism: {{Central imaging review and strict biomarker/tumor tissue handling tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 6 years of follow-up}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W) for IM dosing and every 6 weeks (Q6W) for IV dosing}} **Remote Monitoring Frequency:** {{Between cycles as medically indicated}} **Checklist Review Interval:** {{Imaging review every 9 to 12 weeks for RECIST assessment}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				